MIL-S-20197C
16 March 1977
SUPERSEDING
MIL-S-20197B
29 April 1971

MILITARY SPECIFICATION

SCALE, PLOTTING, PRECISION, CENTIMETERS AND INCHES, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a metal, precision calibrating and measuring scale graduated in centimeters and inches, and a carrying case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The scale shall consist of a metal bar with graduations as shown in Figures 1 and 2, furnished with a case as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.

3.2 First article (first-produced scale). The contractor shall furnish one or more scales as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce scales that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Metal. The scale shall be iron-nickel, low-expansion alloy of 36 percent, plus or minus 1 percent, nickel. Coefficient of expansion shall be 0.000001 to 0.000002 inch per inch per degree C, stress relief at 205° C, plus or minus 5° C, for not less than 48 hours.

3.4 Surfaces and edges. Scale edges shall be square and straight to within 0.005 inch and parallel to within 0.015 inch and shall not vary in thickness by more than 0.003 inch.

3.5 Graduations and numerals. The scale shall be graduated as specified herein and as shown in Figures 1 and 2. One side shall be graduated in centimeters and the other side shall be graduated in inches. All graduations shall be centered approximately in the overall length of the scale and shall be machine-divided. All graduation marks shall be V-shaped, uniform, clear, and distinct, and shall be 0.005 inch, plus or minus 0.002 inch, deep and 0.007 inch, plus or minus 0.002 inch, wide at the top. The error in length between any two graduations shall be not more than 0.002 inch at 72° F (22.2° C). A center punch mark with a depth of not less than 0.005 inch and a 0.015- to 0.018-inch diameter at the top shall be located at the intersection of each vertical line and each vernier slant line with the central horizontal line and at the intersection of each horizontal line with the central vernier slant line. The numbers for the vertical graduations shall be 0.094 inch high. The numbers for the horizontal lines on the inch and centimeter scale shall be 0.062 inch high.

3.5.1 Centimeter scale. The graduations of the centimeter scale shall permit precise measurements to 0.1 of a millimeter (mm), shall extend to the right of the zero mark at 1-centimeter (cm) intervals for a distance of 100 cm, and shall extend to the left of the zero mark for a distance of 1 cm. The centimeter scale shall be in the form of a grid consisting of 11 parallel horizontal lines extending the entire

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.

length of the graduated portion of the scale, spaced 0.125 inch apart, and connected at the ends and at intermediate intervals of 1 cm by lines perpendicular thereto. The centimeter to the left of the zero mark shall be subdivided by 10 slanting, parallel, uniformly spaced lines terminating at the outer horizontal lines of the grid and having such a slope that the upper end of each line shall be to the left of the zero mark at a distance of 1 mm greater than the lower end of the same line. The first sloping line immediately to the left of the zero mark shall coincide with the zero mark at the lower end of the line (at the intersection with the lower horizontal line of the grid) and shall be 1 mm to the left of the zero mark at the upper end (at the intersection with the upper horizontal line of the grid). The next adjacent sloping line shall be 1 mm to the left of the zero mark at the lower end and 2 mm to the left of the zero mark at the upper end. The centimeter graduations for the entire length of the scale and the 5-mm graduations to the left of the zero mark shall be numbered at both the top and the bottom of the grid. The graduations to the right of the zero mark shall read from left to right and the graduations to the left of the zero mark shall read from right to left. The intermediate horizontal lines on the grid shall be numbered at both ends of the graduations from bottom to top from 1 to 9, inclusive.

3.5.2 Inch scale. The graduations of the inch scale shall permit precise measurements to 0.01 of an inch, shall extend to the right of the zero mark at 1-inch intervals for a distance of 39 inches, and shall extend to the left of the zero mark for a distance of 1 inch. The graduation of the inch scale shall be identical to the graduation of the centimeter scale as specified in 3.5.1 except that the divisions shall be in inches and in tenths and hundredths of an inch.

3.6 Identification marking. Each scale shall be identified in accordance with MIL-STD-130.

3.7 Case. Each scale shall be contained in a felt-lined wooden case with a hinged lid. The base and lid of the case shall be made in such a manner that the centimeter and inch scales can be used without removing or turning the scale.

3.8 Workmanship. The scale shall be clean and free from burrs and sharp edges.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:24Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced scale inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced scale inspection.

4.3.1 Examination. The first-produced scale shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first-produced scale shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be conducted in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:24Z
Check the source to verify that this is the currentversion before use.

4.4.3 Test.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The scale shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Edges not square, straight, or parallel, and dimensions not within specified limits.
103. Scales not graduated as specified.
104. Case not as specified.
105. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.5.2 Accuracy test. A beam compass which has been certified as accurate by the National Bureau of Standards shall be used to check the distances between any two graduations. An error in length greater than 0.002 inch between any two graduations on the rule at 72° F (22.2° C) shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:24Z
Check the source to verify that this is the currentversion before use.

106. Materials, method, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
107. Scale not contained in the carrying case as specified.
108. Box closure not Method V as specified.
109. Strapping not as specified.
110. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each scale, in the carrying case, shall be preserved in accordance with MIL-P-116, Method III in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each scale, in the carrying case, shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The scales, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified for Level A, except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. The scales, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:24Z
Check the source to verify that this is the currentversion before use.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Commercial packaging marking shall reflect as a ~~minimum~~, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The scales are intended for use as standard or master scales for calibrating other equipment, or for making or transferring accurate measurement by a beam compass, or similar device, in map projection work.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced scale, and number of scales required (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When other than Finish B strapping is required (see 5.2.1).

6.3 First-produced scale. Any changes or deviations of production scales from the approved first-produced scale during production will be subject to the approval of the contracting officer. Approval of the first-produced scale will not relieve the contractor of his obligation to furnish scales conforming to this specification.

6.4 It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

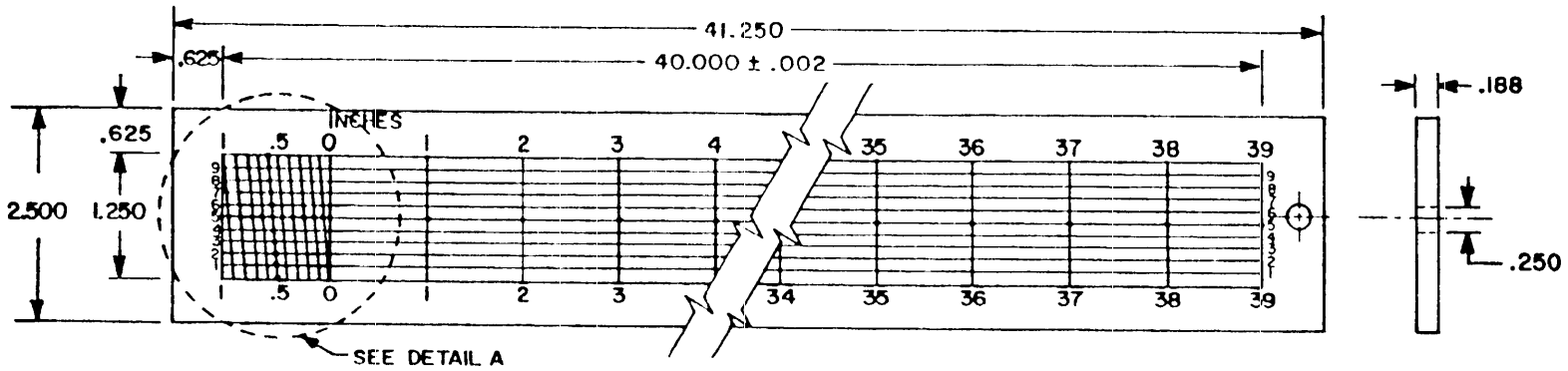
Custodians:
Army - ME
Air Force - 99

Review activity:
DSA - GS

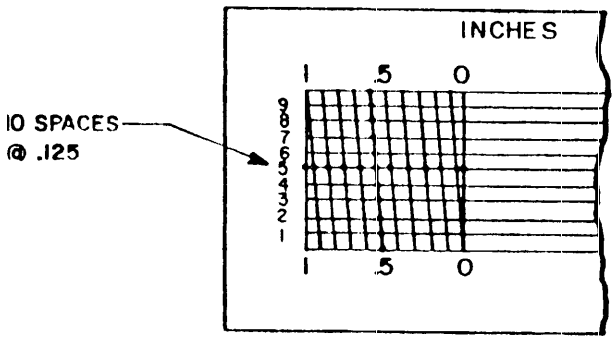
Preparing activity:
Army - ME

Project 6675-0635

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.



BOTTOM VIEW



DETAIL A

INCHES	MM
.002	.05
.125	3.18
.188	4.76
.250	6.36
.625	15.88
1.250	31.75
2.500	63.50
40.000	1016.00
41.250	1047.75

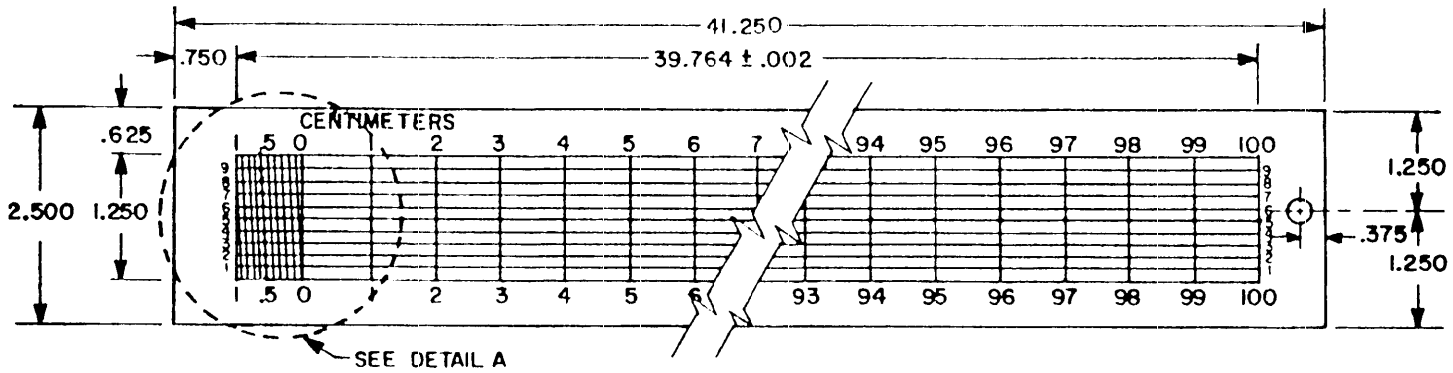
NOTES:

1. ALL DIMENSIONS ARE IN INCHES (MILLIMETERS) SEE TABLE.
2. METRIC DIMENSIONS ARE REFERENCE ONLY.

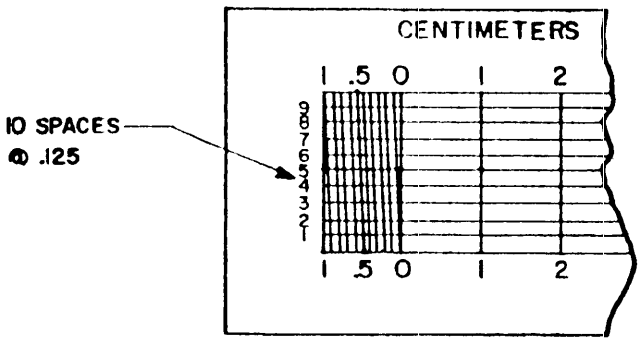
FIGURE 1. Scale plotting graduated in inches.

X-3226

Source: https://assist.dla.mil -- Downloaded: 202640-19T07:24Z
Check the source to verify that this is the current version before use.



TOP VIEW



DETAIL A

INCHES	MM
.002	.05
.125	3.18
.375	9.53
.625	15.88
.750	19.05
1.250	31.75
2.500	63.50
39.764	1010.01
41.250	1047.75

NOTES:

1. ALL DIMENSIONS ARE IN INCHES (MILLIMETERS) SEE TABLE.
2. METRIC DIMENSIONS ARE REFERENCE ONLY.

FIGURE 2. Scale plotting graduated in centimeters.

X-3227

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-S-20197C Scale, Plotting, Precision, Centimeters and Inches, with Case

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:24Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:24Z
Check the source to verify that this is the current version before use.